

Reproducibility Trace Report

Project: Synthetic Primary-Crusher Energy Pipeline

Generated: 2 August 2026

Scientific status: synthetic demonstration only

Execution trace

Step	Command	Output
1	python src/generate_data.py	2,625 synthetic rows
2	python src/train.py	Model, predictions, metrics
3	python src/evaluate.py	Chart and summary

Synthetic temporal-holdout metrics

Model	MAE	RMSE	R2
historical_mean	283.34	359.62	-0.002
linear_regression	106.00	135.67	0.857
hist_gradient_boosting	121.37	154.23	0.816

Artifact identity

params.yaml

SHA-256: 03e0e4a59fbace0f263055a2aab309cffc59eff2c07da3911f468559e3ea0045

Bytes: 656

data/processed/synthetic_blast_crusher.csv

SHA-256: e9d11c50a20efb558f796a7c17adc981ecd167af5736e89d76b430e8ec9038c0

Bytes: 644813

results/metrics.json

SHA-256: 2c536ecb5add181872b802b6f5644c958cb7bcbe4d20c5acea469745bedac49b

Bytes: 623

results/model.joblib

SHA-256: 9f149047e679fbe96a38dce79b5615236683f9086fdd295814985f944061c27e

Bytes: 925874

Integrity statement

This report proves execution and traceability for the public synthetic pipeline. It does not prove predictive validity on the confidential operational dataset.